Test Case 1.1 (complex test)

Test Case #: 1.1	**Test Case Name:** Student Sign-in	**Page:** 1 of 1
System: UniCore	**Subsystem:** Authentication	
Designed by: Manan Kapoor	**Design Date:** 02/11/2026	
Executed by:	**Execution Date:**	

Short Description: Test that a student can sign in to UniCore, and that invalid credentials are rejected

Pre-conditions

The UniCore website is reachable at `https://abhinav29102005.github.io/unicore/sign`

A student account `abhinav@unicore.edu` exists with status *active* and the role *Student*

The browser is not signed in to UniCore (site data cleared or a private window is used)

API base URL, written as `<API>` below: `https://unicore-backend.bigboyaks-account.workers.dev/api`

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Open the sign-in page URL	The system displays the sign-in page with the roles Student, Faculty, Admin and Staff. *Student* is selected and the card shows "Abhinav Kumar – Active Student"		
2	Open the browser DevTools → Network tab	The Network tab is open and recording requests		
3	Click the 'Continue as Abhinav (Student)' button	The request `POST /api/auth/login` returns status **200** with the user "Abhinav Kumar", role "Student" and a token. The system opens the student dashboard (`/unicore/student`)		
4	**Check post-condition 1**			
5	Send a login request with a **wrong password**: `curl -X POST <API>/auth/login -H "Content-Type: application/json" -d '{"email":"abhinav@unicore.edu", "password":"wrong-password"}'`	The system responds with status **401** and the message "Invalid credentials". No token is returned		
6	Repeat step 5 with an email that does not exist, say `nobody@unicore.edu`	The system responds with status **401** and the message "Invalid credentials". No token is returned		

Post-conditions

1. The last login time of `abhinav@unicore.edu` is updated in the database. Check by opening `<API>/auth/profile/7b669c2a40a0403f785eccd291f51b6a`: `lastLogin` shows the time of step 3 (UTC)

Tests the happy path (steps 1–4) and failure conditions (steps 5–6).

Test Case 1.2 (complex test)

Test Case #: 1.2	**Test Case Name:** Student Dashboard	**Page:** 1 of 1
System: UniCore	**Subsystem:** Student Portal	
Designed by: Manan Kapoor	**Design Date:** 02/11/2026	
Executed by:	**Execution Date:**	

Short Description: Test that the student dashboard shows the signed-in student's live data

Pre-conditions

Test Case 1.1 steps 1–3 are completed: the user is signed in as student Abhinav Kumar (roll no. 1024030440)
The student has 4 course enrollments, an active hostel allocation and 1 issued library book in the database
The browser DevTools → Network tab is open
API base URL, written as `<API>` below: `https://unicore-backend.bigboyaks-account.workers.dev/api`

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Open the student dashboard (`/unicore/student`)	The system displays "Welcome back, Abhinav" and "1024030440 | Computer Science & Engineering, Year 3"		
2	In the Network tab, select the request `/api/student/1024030440/dashboard`	The request returned status **200** with the student's data (this confirms the page shows live data, not the offline fallback)		
3	Check the 'Current CGPA' card	The system displays CGPA **8.13**		
4	Check the 'Hostel Status' card	The system displays "Tagore Hall - 301" with status *active*		
5	Check the 'Recent Enrollments' section	The system lists 4 courses: CS301 Distributed Systems (B+), CS302 Cloud Computing (A+), CS303 Database Management Systems (B+), CS304 Machine Learning (B)		
6	Check the 'Library Assets' section	The system lists "Designing Data-Intensive Applications" with its due date		
7	Open the dashboard API for a roll number that does not exist: `<API>/student/0000000000/dashboard`	The system responds with status **404** and the message "Student not found"		
8	**Check post-condition 1**			

Post-conditions

1. No data is changed in the database (the dashboard only reads data)

Tests the happy path (steps 1–6) and a failure condition (step 7).

Test Case 1.3 (complex test)

Test Case #: 1.3	**Test Case Name:** Student Attendance	**Page:** 1 of 1
System: UniCore	**Subsystem:** Academic – Attendance	
Designed by: Manan Kapoor	**Design Date:** 02/11/2026	
Executed by:	**Execution Date:**	

Short Description: Test that the attendance service returns correct per-course attendance and percentages

Pre-conditions

Student 1024030440 is enrolled in CS301, CS302, CS303 and CS304

Each course has 5 attendance records for the student in the database

API base URL, written as `<API>` below: `https://unicore-backend.bigboyaks-account.workers.dev/api`

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Open `<API>/student/1024030440/attendance` in the browser	The system responds with status **200** and an `attendance` list containing 4 courses		
2	Check the CS301 Distributed Systems entry	present = 3, absent = 2, total = 5, percentage = **60**		
3	Check the CS302 and CS303 entries	Each shows present = 4, absent = 1, total = 5, percentage = **80**		
4	Check the CS304 Machine Learning entry	present = 5, absent = 0, total = 5, percentage = **100**		
5	For every course, calculate (present + late) ÷ total × 100	The result equals the `percentage` shown by the system		
6	Repeat step 1 with a roll number that does not exist, say `0000000000`	The system responds with status **404** and the message “Student not found”		
7	**Check post-condition 1**			

Post-conditions

1. No data is changed in the database (attendance is only read)

Tests the happy path (steps 1–5) and a failure condition (step 6).